

Vikranth Chapaala

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SUMMARY

AI Engineering and Data Professional with 4+ years of experience in delivering end-to-end, production grade Generative and Agentic AI solutions aligned to business goals. Skilled in designing scalable RAG pipelines and multi-agent workflows, with deep expertise in retrieval, guardrails, and evaluation frameworks across cloud-native environments.

PROFESSIONAL EXPERIENCE

Data Scientist/AI Engineer - Paycom | Fulltime, Dallas, TX

May 2024 -Current

- Architected and deployed 6+ Agentic AI workflows using LLM-based reasoning, tool execution, and enterprise data integrations across different domains, automating complex operational processes and driving an estimated **\$6.6M annual business impact**.
- Owned delivery and implementation of an enterprise RAG framework powering downstream applications including Help Center search and conversational knowledge assistants, leveraging document ingestion pipelines, embedding generation, semantic retrieval, which improved knowledge discovery efficiency and reduced employee **support queries by 32%**.
- Developed a unified AI data layer combining knowledge graphs (Neo4j), vector search (Elasticsearch), and multi-agent workflows built with OpenAI SDK and LangChain to process structured and unstructured enterprise data, enabling AI-driven reporting and contextual decision support.
- Built an LLM evaluation and governance framework (LLMEvalKit) supporting **28+ evaluation metrics across 13 modules** for RAG pipelines, agentic AI systems, chatbots, and document intelligence applications; enabled automated quality assessment, hallucination detection, and AI reliability testing.
- Established an enterprise LLM security and guardrails layer combining model-based safety checks, deterministic policy engines, and automated validation pipelines to govern AI behavior **across 10+ systems**, multi-agent workflows, and production LLM applications.
- Containerized, deployed, and managed cloud-native AI applications on Kubernetes clusters across cloud environments, supporting scalable LLM inference deployments, production workloads, and end-to-end model lifecycle management.
- Contributed to AI observability and tracing infrastructure across production LLM pipelines, monitoring latency, cost, and output quality to drive proactive model performance tuning and **cost optimization strategy**.

Data Science Engineer – Brillio | Fulltime, Remote

Sep 2021-Jul 2023

- Played a key role to develop a modern Data layer from scratch, **saving the client \$2.6M** and enabling data science teams to access clean, real-time data for modeling and analysis.
- Automated end-to-end data pipelines for ingesting and transforming multi-source data, cutting manual processing effort **by 2%** and improving data quality for downstream teams.
- Built automated dashboards tracking campaign performance and customer engagement KPIs, reducing reporting turnaround from weeks to days for business stakeholders.

Business Systems Analyst – iCompaas | Fulltime, Remote

Aug 2020-Aug 2021

- Conducted thorough analysis of business requirements, collaborated with cross-functional teams to drive process improvements in the product reporting and management to enhance the data accuracy.
- Managed offshore operations analytics, including planning, tracking, and Tableau-driven data reporting, achieving an 8% enhancement in product usage metric monitoring and reporting effectiveness.
- Supervised a comprehensive knowledge hub covering 12+ AWS services, imparting invaluable technical guidance to end- users.

TECHNICAL SKILLS

Generative AI & Agentic AI: OpenAI SDK, LangChain, LangGraph, MCP, RAG, Multi-Agent Orchestration, Agentic Workflows, LLM Fine-Tuning (LoRA/QLoRA), OCR, Prompt Engineering, Function Calling, Tool Use, LLM Evaluation, Guardrails, Observability

Cloud & MLOps: AWS, Azure, vLLM, LiteLLM, Docker, Kubernetes, CI/CD Pipelines, Model Deployments, Model Monitoring

Programming & Data Engineering: Python (Pandas, NumPy, Scikit-learn, PyTorch, TensorFlow, Keras), SQL, PySpark, FastAPI, REST APIs, Data Pipelines, Structured & Unstructured Data Processing

LLM Optimization & Inference: Inference Engine Benchmarking, Throughput & Latency Optimization, Quantization

EDUCATION

- MS in Business Analytics & Data Science | Oklahoma State University, USA | GPA: 3.9/4
- Bachelor's in Computer Science Engineering | Dr. MGR University, India | GPA: 8.8/10

May 2025
May 2021